
Muchamad Fikri Ali

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A Computer Science undergraduate at Muhammadiyah University of Cirebon with a primary focus on Internet of Things (IoT) development and a strong interest in web development. Experienced in designing and developing IoT-based systems, including microcontroller programming, sensor integration, communication protocols, and hardware-software integration. Also skilled in front-end and back-end web development, API integration, and automation scripting. Able to work effectively both independently and as part of a team, with strong communication and problem-solving skills. Passionate about building practical, reliable, and innovative technology solutions while continuously developing technical expertise.

Education

Muhammadiyah University of Cirebon

Cirebon, Indonesia

Bachelor of Informatics Engineering.

October 2022 – October 2026

Undergraduate Thesis: Design and Development of an Accident Detection System and Blackbox for 2-Wheel Vehicles Based on The Internet of Things.

MA Al-hikmah 2 Cirebon Cirebon, Indonesia

Cirebon, Indonesia

Natural Science Major (IPA) • Final Score: 86.8/100

2021

2021 Academic Honors: Silver Medalist in the National Indonesian Language Competition.

Technical & Project Experience

Design and Development of an IoT-Based

Cirebon, Indonesia

Accident Detection and Black Box System for Two-Wheeled Vehicles

Maret 2026 – July 2026

Independent / Academic Project

- Designed and developed an accident detection and black box system for two-wheeled vehicles based on the Internet of Things (IoT).
- Integrate sensors, microcontrollers, and IoT communication systems to detect accidents and record and transmit vehicle data in real time.

IoT-Based Automatic Plant Watering System

Cirebon-Yogya, Indonesia

UGM “Kampus Merdeka” (Wirausaha Merdeka) IoT Development Project

August 2024–December 2024

- Develop an IoT-based automatic plant watering system with three operating modes: automatic, scheduled, and manual.
- Integrated soil moisture sensors, a microcontroller, a water pump, and IoT connectivity to monitor soil conditions and control the watering process efficiently.

IoT-Based Hydroponic Monitoring System

Cirebon, Indonesia

Independent IoT Development Project

June 2026 – Agustus 2026

- Developed an IoT-based hydroponic monitoring system to monitor water temperature, pH levels, and nutrient conditions in real time.
- Integrated sensors and microcontrollers with a monitoring interface to enable real-time data collection and improve the efficiency of hydroponic management.

Cashier POS Application

Cirebon, Indonesia

Independent Software Development Project

June 2026 – September 2026

- Developed a Point of Sale (POS) application to manage product data, sales transactions, payment processes, and transaction history.
- Implemented a structured database and a user-friendly interface to streamline transaction processes and improve cashier operational efficiency.

PWS KIA Mother and Child Clinic Apps – Front-End Developer

Cirebon, Indonesia

Front-End Development Project

July 2024 – September 2024

- Developed and implemented a responsive front-end interface for the PWS KIA (Local Monitoring of Maternal and Child Health) Mother and Child Clinic app.
- Implemented UI/UX designs into functional web interfaces and optimized the layout to ensure it is responsive, user-friendly, and compatible across various devices.

Leadership & Activities

COMPUTER SCIENCE STUDENT ASSOCIATION

Cirebon, Indonesia

Head of the Interests and Talents Division

November 2022 – December 2024

- Manage the work program for the student interests and talents division and develop activities that support members' potential, creativity, and skills.
- Coordinate the implementation of activities, assign tasks to members, and ensure the division's work program runs effectively.

Al-Hikmah Student Activities Forum (FASIH)

Cirebon, Indonesia

Chief Secretary

July 2020 – July 2021

- Managed organizational administration, document preparation, correspondence, and archiving of organizational activities.
- Coordinated administrative needs and supported the smooth implementation of various FASIH work programs.

National Event Competition for Islamic Student (CIS)

Cirebon, Indonesia

Chief Secretary

Desember 2020 – April 2021

- Supervised daily musical and marching drills, structured practice routines, and maintained team performance standards.

Certifications

- Junior Web Developer – National Board for Professional Certification (BNSP) (2025) | No. Reg. TIK 1565 37377 2025
- Junior Web Developer – Vocational School Graduate Academy Digital Talent Scholarship (VSGA)

Skills & Interests

Technical Skills: Internet of Things (IoT), Embedded Systems, Microcontroller Programming, Full-Stack Web Development, Database Management, Automation Scripting.

Soft Skills & Competencies: Technical Communication, Complex Problem-Solving, Cross-Functional Team Leadership, Project Coordination.

Hardware & Embedded System: C++, Microcontroller Programming, Arduino, ESP8266, ESP32, ESP32-CAM, Robotics, Circuit & Sensor Interfacing, electronic and electrical hardware.

Web Development: Full-Stack Web Development, HTML, CSS, JavaScript, PHP, Laravel, React, Vite.

Data, AI & Databases: Python, TensorFlow, YOLO (Object Detection), MySQL, Firebase.

Tools & Infrastructure: GitHub, Vercel, Laragon, Postman, MQTT Protocol.

Soft Skills & Competencies: Technical Communication, Complex Problem-Solving, Team Collaboration, Organizational Leadership.

Language: Bahasa Indonesia

Interests: Robotics, Embedded Systems, Internet of Things (IoT) Innovation.